Ontologies, Multi-agent Digital Twin and Intelligent Telemetry for the Internet Infrastructure (DT-II)

Juliao Braga
Center for Mathematics, Computation and Cognition
Federal University of ABC
Santo André, SP, BR
`juliao.braga@ufabc.edu.br`

Jéferson Campos Nobre
Institute of Informatics - PPGC
Federal University of Rio Grande do Sul
Porto Alegre, RS, BR
`jcnobre@inf.ufrgs.br`

Juliana C. Braga
Center for Mathematics, Computation and Cognition
Federal University of ABC
Santo André, SP, BR
`juliana.braga@ufabc.edu.br`

Itana Stiubiener
Center for Mathematics, Computation and Cognition
Federal University of ABC
Santo André, SP, BR
`itana.stiubiener@ufabc.edu.br`

Leandro Marcio Bertholdo
Institute of Informatics - PPGC
Federal University of Rio Grande do Sul
Porto Alegre, RS, BR
`leandro.bertholdo@gmail.com`

Marcelo Santos
Institute of Informatics
IFSertão Pernambucano
Salgueiro, PE, BR
`marcelo.santos@ifsertao-pe.edu.br`

Percival Henriques
Brazilian Internet Steering Committee
São Paulo, SP, BR
`percival@cgi.br`

Abstract

The Internet Infrastructure is dynamic and heterogeneous, which makes its governance difficult through manual procedures and vendor-specific models. This paper presents DT-II, an ontology-driven Digital Twin designed to represent, monitor, and govern this infrastructure semantically. The proposal defines an Upper Ontology, grounded in the Unified Foundational Ontology (UFO) and modelled in OntoUML, centred on the triad `Node-Connection-Channel`. The model is formalised in Description Logic and operationalised as a Knowledge Graph, whose instance base (ABox) is continuously synchronised via model-driven telemetry (e.g., YANG-Push). On this basis, DT-II applies automated reasoning (TBox/RBox/ABox) to support semantic interoperability, fault detection/diagnosis, and compliance checks (e.g., SLAs and Ontological Compliance Regime). Finally, the system is organised as a multi-agent architecture based on MAPE-K, enabling explainable and policy-driven actions across heterogeneous network domains.

Keywords Domain Ontology · Foundational Ontology · UFO · OntoUML · NETCONF · Multi-agent Digital Twin

1 Introduction

The Internet Infrastructure (II) has become **highly** dynamic and heterogeneous, making it unreliable to govern solely through manual procedures, *ad hoc* scripts, and vendor-specific interpretations. In this context, ***Knowledge Graphs*** (KGs) have been employed to represent the operational state as connected facts, whilst ***ontologies*** provide the vocabulary, constraints, and consistency criteria that render these facts interpretable and verifiable (Kejriwal, Knoblock, and Szekely 2021; Landgrebe and Smith 2025; Juliao Braga, Henriques, et al. 2025). In DT-II, the ontology defines the logical framework (classes, relations, and axioms), and the KG represents its instantiation, continuously updated.

This paper proposes a ***Upper Ontology*** (UO) for II (Arp, Smith, and Spear 2015), grounded in the Unified Foundational Ontology (UFO) and formalised in Description Logic (DL) (Baader et al. 2017; Nardi and Brachman 2010; Horrocks, Kutz, and Sattler 2006). In practical terms, the UO acts as a **Normative Ontological Basis**: it establishes core concepts and common rules to unify heterogeneous subdomains (e.g., 5G, satellite, and optical) around the triad **Node–Connection–Channel**. Modelling in OntoUML/OLED separates *types* bearing identity (Kind) from *functions* dependent on context (Role), which favours systematic validation and export to OWL/Turtle (Guerson et al. 2015; Guizzardi, Fonseca, et al. 2018; W3C OWL Working Group 2012). For reference throughout the text: ***TBox*** describes the conceptual schema, ***RBox*** describes relational constraints/axioms, and ***ABox*** describes instances (the observed state of the network).

1.1 Contributions

The main contributions of this work are:

- an ontology-driven reference model for II centred on *Node–Connection–Channel*, grounded in UFO and developed in OntoUML;
- an operationalisation in DL/OWL supporting automated reasoning over ***TBox***, ***RBox***, and ***ABox***;
- a synchronisation approach aligning the KG with model-driven telemetry (e.g., YANG-Push) for near real-time state updates;
- a governance mechanism based on an **Ontological Compliance Regime** and SLA constraints, enabling auditable and explainable compliance checks; and
- a multi-agent Digital Twin architecture following the ***MAPE-K*** cycle (Kephart and Chess 2003; IBM Corporation 2005) for policy-driven actions.

1.2 Ontological Compliance Regime (overview)

The Ontological Compliance Regime is a normative layer (in other words, a governance mechanism) defined by ontological axioms and mediated by relators (Guizzardi 2005; Almeida, Guizzardi, Tiago P. Sales, et al. 2019; Almeida, Guizzardi, Tiago Prince Sales, et al. 2026). It enables the Digital Twin to determine which constraints (e.g., legislation, SLAs, privacy policies) apply to entities and data flows, and to propagate these constraints across nodes, connections, and channels through automated reasoning.

The main characteristics and functions of this regime include:

- **Automated Propagation:** By employing a logical reasoner, the regime propagates normative constraints across nodes, connections, and network channels. If a node is bound to a specific legal norm, all connections and channels mediated by it inherit those same constraints.
- **Jurisdiction Redefinition:** In this model, jurisdiction ceases to be a mere static hardware attribute delimited by physical geography, and is instead understood as a normative entity based on meaning, context, and relations.
- **Legally Aware Infrastructure:** The regime transforms the infrastructure into a "legally aware" system, which means that the compliance of network operations is mathematically verified through DL.
- **Auditability and Explainability:** The regime ensures that compliance checks are both auditable and explainable. In dynamic events, such as failures or attacks, decisions taken by the MAPE-K cycle are only executed in the network after being logically validated by this normative base.

1.3 A practical roadmap for the DT-II project

DT-II is the name assigned to this project. This section describes its various components in detail. Figure 1 summarises the project's **Intelligence Cascade**.

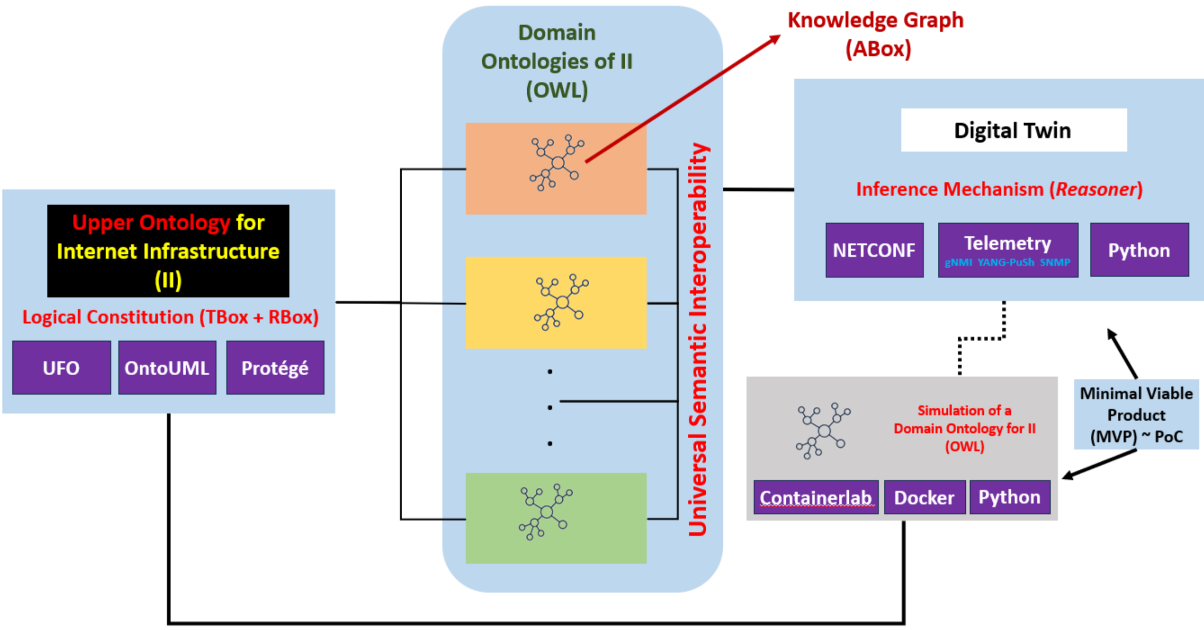

Figure 1: Overview of the DT-II Project Architecture

The figure illustrates how the UO, DOs, and the Digital Twin interact to ensure semantic interoperability[1] (Guizzardi and Guarino 2024) and to promote dynamic governance. Furthermore, it characterises the testbed environment and the *Minimum Viable Product* (MVP), in this case representing the Proof of Concept (PoC).

On 2026-07-06, the Internet had approximately **78,964** active Autonomous Systems[2] (ASs) (Hawkinson and Bates 1996), as reported in the **CIDR IPv6 Report**[3] maintained by Geoff Huston, Chief Scientist from Asia-Pacific Network Information Centre (APNIC). Each AS possesses operational autonomy, with its functionality established by protocols and specialised administrators. The correctness of this functionality characterises the behaviour of the Internet from the user's perspective. A misconfiguration in one AS can propagate across the global II, causing worldwide incidents. For example, if an AS incorrectly announces an IPv4 block via the BGP protocol, it may cause serious damage, particularly to the legitimate holder of that block. One way to prevent anomalous behaviours in the AS environment is through measurements known as **telemetries**.

These risks highlight the need for advanced monitoring mechanisms. The DT-II project proposes the simulation of ontology-based techniques (or knowledge bases) capable of establishing intelligent telemetries without human intervention. Alongside ontologies, protocols such as NETCONF and related technologies are employed, widely deployed and implemented by major II equipment vendors.

The project employs two types of ontologies: domain ontologies (DOs), which specify the knowledge of each AS, and the UO, which ensures semantic interoperability and facilitates the construction and verification of DOs through the OntoUML tool. Although DOs are similar across most ASs, the innovation of the project lies in the use of the UO, which brings benefits such as:

- Ensuring semantic interoperability across different domains;
- Facilitating systematic construction and verification of DOs via OntoUML.

Additionally, the project uses *Protégé*, supported by gUFO, for modelling DOs both in the testbed and in ASs participating in the second phase of the research. The testbed, developed with ***Containerlab***, ***Docker***, and ***Python***, constitutes the DT-II PoC, enabling validation of the project's implementation.

[1] Semantic interoperability refers to the ability to connect different information systems with precise understanding of how real-world entities represented in these diverse systems relate to each other, going beyond mere packet exchange (technical interoperability) or data format comprehension (syntactic interoperability).

[2] An Autonomous System (AS) is the fundamental unit of routing on the Internet, defined as a set of IP networks under a single routing policy and administration.

[3] `https://www.cidr-report.org/`

The DT is the operational component of the project. It adopts the autonomic characteristics described in Richard Murch's book "Autonomic Computing" (Murch 2004) and is expected to behave in accordance with the eight elements proposed to preserve the characteristics of an autonomic computing system:

- Comprehensive self-knowledge, including available resources and operational capabilities
- Ability to configure and reconfigure itself autonomously in response to contextual requirements
- Capacity for continuous self-optimisation, ensuring sustained efficiency and performance
- Robust self-healing mechanisms to detect, isolate, and recover from failures
- Self-protection capabilities to safeguard against internal inconsistencies and external threats
- Ability to acquire and interpret knowledge of its environment and context, adapting behaviour accordingly
- Competence to operate effectively in heterogeneous and distributed computing environments
- Ability to anticipate, interpret, and adapt to evolving user requirements

A DT can maintain an environment that preserves the above elements (e.g., self-knowledge, self-configuration, self-optimisation, self-healing, and self-protection) if conceived as a **multi-agent** architecture supported by a semantic knowledge base. To this end, the DT follows the **MAPE-K** proposal (Kephart and Chess 2003; IBM Corporation 2005), which establishes the following cycle as a means of ensuring autonomy:

- **Monitoring (M)**: collects real-time data.
- **Analysis (A)**: interprets the data based on axioms and rules.
- **Planning (P)**: generates action plans consistent with the Normative Ontological Basis.
- **Execution (E)**: applies changes to the infrastructure.
- **Knowledge (K)**: feeds back into the knowledge base, ensuring continuous learning.

Figure 2 provides a detailed visual representation of the MAPE-K flow of the DT proposed in this article.

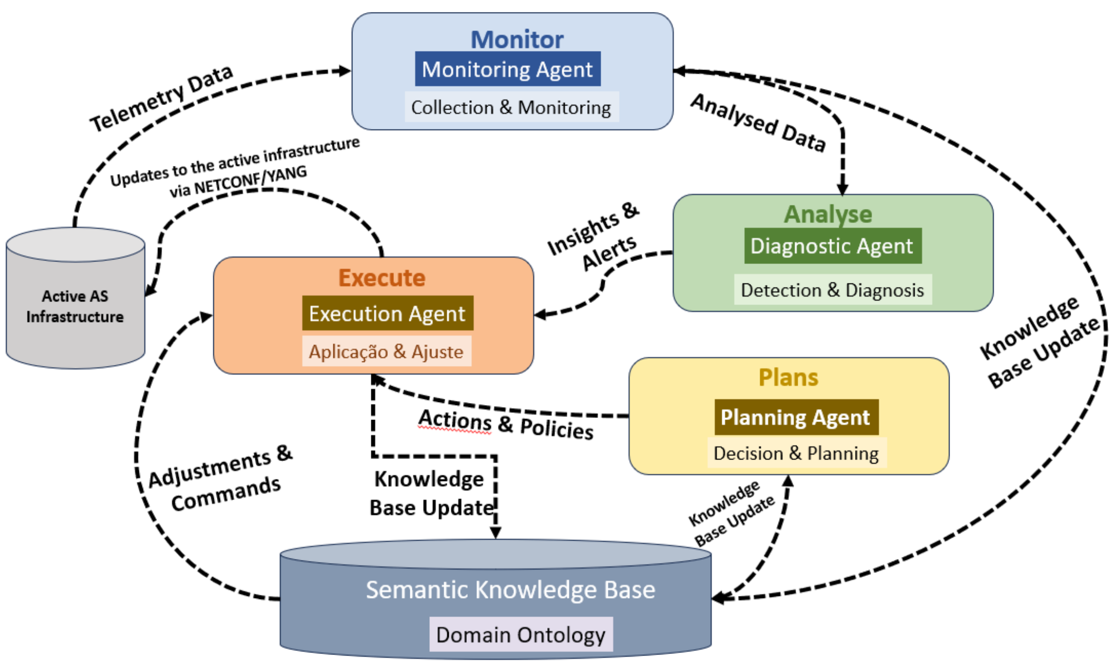

Figure 2: MAPE-K Flow of the Digital Twin

2 Related Work

This section positions DT-II in relation to three research strands: (i) **network digital twins** and closed-loop/autonomic management; (ii) **ontology-based network modelling** and knowledge graphs for inventory, topology, and intent; (iii) **telemetry-driven inference** and policy/compliance enforcement.

2.1 Network Digital Twins and Autonomic Control

Digital Twins provide a digital counterpart of an operational system, enabling monitoring, diagnosis, and actuation (Grieves and Vickers 2017). In networking, recent proposals emphasise closed-loop control, model-driven operations, and integration with telemetry (Zhou et al. 2026). DT-II differs by making the twin *explicitly governed* by a superior Normative Ontological Basis, so that actions are validated against semantics and constraints rather than solely metrics or learned policies.

2.2 Ontologies and Knowledge Graphs for Network Infrastructure

Knowledge graphs have been applied to integrate heterogeneous network data sources and to support inter-domain queries such as in Thomas et al. (2026). Ontology-driven approaches add explicit meaning, identity criteria, and constraints, enabling consistency checking and explainable inference. DT-II builds upon this line by grounding its central model in UFO (via OntoUML/gUFO), explicitly modelling roles and relators, and using this structure as a foundation for interoperability across multiple administrative domains.

2.3 Telemetry, Model-driven Interfaces, and Reasoning

Modern network management increasingly relies on streaming telemetry and model-driven interfaces (e.g., YANG-based mechanisms) to maintain near real-time state. DT-II adopts model-driven telemetry via YANG-Push (RFC 8641) (Clemm and Voit 2019), as its synchronisation mechanism, combining it with DL reasoning over ***TBox*/*RBox*/*ABox*** to detect inconsistencies and to derive governance-relevant states (e.g., SLA violations, jurisdictional conflicts).

2.4 Governance, Policies, and Compliance

Whilst policy frameworks typically impose restrictions through configuration templates or procedural checks, DT-II formalises governance as part of the ontology itself (e.g., Ontological Compliance Regime, SLA relators). This enables automated validation, traceability, and explainability of compliance decisions across domains.

3 Methodology

This project follows the ontology engineering lifecycle (Keet 2025; Parks 2025; Effingham 2013), employing the ***Unified Foundational Ontology*** (UFO) as its basis and OntoUML as the modelling language (Guizzardi 2005). The process is structured into four technical phases:

M1. **Conceptual Abstraction**: UFO theory is applied to classify network assets. Rigid essence classes (`PhysicalElement`, `MaterialMedium`) define identity, whilst accidental roles (`NetworkNode`, `CommunicationChannel`) are linked through relators.

M2. **Formalisation in Description Logic (DL)**: Concepts are axiomatized in DL, ensuring computational interpretability. For example, every `NetworkNode` must possess a logical identifier:

$$\text{NetworkNode} \sqsubseteq \text{PhysicalElement} \sqcap \exists \text{possesses}.\text{LogicalIdentifier} \tag{1}$$

M3. **Semantic Serialisation**: Models validated in ***OLED*** are exported to OWL and serialised in ***Turtle***. Standard vocabularies such as ***Dublin Core*** (DC) and ***FOAF*** provide support for metadata and provenance.

M4. **Automated Reasoning**: Inference engines confront the ***TBox*** and the ***ABox*** to verify consistency and detect anomalies. Predicate logic enables deduction of failure states; for instance, an inactive channel implies compromise of its logical connection.

3.1 Ontological Analysis Structure and Mapping Process

The mapping between the II domain and UFO, via its gUFO implementation, follows a specialisation protocol grounded in logical metaproperties. For each local domain entity, the **Ontological Analysis Structure** is applied (Almeida, Guizzardi, Tiago Prince Sales, et al. 2026; Almeida, Guizzardi, Tiago P. Sales, et al. 2019; Guizzardi 2005):

A. **Specialisation of Fundamental Types**: Entities with rigid identity and specific individuation principles (e.g., Physical Routers, Satellites, Fibre Cables) are mapped as subclasses of *gufo:Kind*. These represent substantial objects that persist independently of their configuration state.

B. **Relational Specialisation (Roles)**: Concepts whose existence depends on a specific topological or functional context (e.g., Edge Node, Peering Point, Transit AS) are characterised as *gufo:Role*. This allows the Digital Twin to capture the dynamic nature of network functions independently of the underlying hardware.

C. **Relational Foundations**: Logical connections and service agreements establishing dependencies between multiple objects (e.g., BGP Sessions, SLA Agreements) are formalised as *gufo:Relator*. This ensures that the integrity of the Knowledge Graph is maintained through existential dependency constraints inherent to gUFO, where invalidation of a relator triggers logical removal of the associated roles.

By employing this structure, DT-II transcends mere data modelling, establishing a **Normative Ontological Basis** in which network behaviour is governed by foundational ontological axioms rather than purely heuristic or statistical patterns.

3.2 Dynamic Synchronisation via YANG-Push

A critical requirement for DT-II is high-fidelity synchronisation between the physical infrastructure and its semantic representation. To achieve this, the model employs *Model-Driven Telemetry* (MDT) through the **YANG-Push** mechanism (RFC 8641 (Clemm and Voit 2019)).

Unlike traditional query-based monitoring (*pull*), YANG-Push enables Network Nodes to directly transmit state changes to the *Semantic Ingestor*. This integration is mapped into the gUFO framework as follows:

- **Periodic Subscriptions**: Quantitative metrics, such as interface counters, are transmitted to update `gufo:Quality` assertions in the ABox.
- **Change Notifications**: Critical events, such as oscillations in BGP sessions, are treated as `gufo:Events` that immediately trigger re-evaluation of existential dependencies in `gufo:Relators`.

This ensures that the Normative Ontological Basis of the Digital Twin always reflects the current operational state of ASs with latency below one second.

4 Relations between Ontology, Knowledge Graph, and Description Logic

The ***Knowledge Graph*** (KG) operationalises the ontology by linking **real data** (instances) to the conceptual model. It represents information as **nodes** (entities) connected by **edges** (relations). For example, an instance `inst:Router_R1` may be linked to `inst:Router_R2` via `ii:connectedTo`, and the resulting `ii:Connection` may be linked to an `ii:Channel` with capacity and latency qualities. Typically, a KG integrates multiple sources to form a network of interconnected facts.

Ontology and KG are **complementary**: the ontology provides semantics and logical rules, whilst the KG instantiates them as a dynamic and queryable knowledge base. Table 1 summarises their roles in the proposed framework.

Table 1: Complementarity between Ontology and Knowledge Graph

Characteristic	Ontology	Knowledge Graph
Focus	Definitions, rules, logic.	Data, facts, connections.
Level	Abstract (the “DNA”).	Concrete (the “Organism”).
Function	Provides meaning and semantics.	Enables queries and discoveries.

Description Logic (DL) underpins modern ontologies (e.g., OWL in the Semantic Web) (Baader et al. 2017). DL enables the formal definition of concepts and computational reasoning. For example, the axiom:

$$\text{NetworkNode} \sqsubseteq \text{PhysicalElement}$$

establishes that every `NetworkNode` is a `PhysicalElement`. More complex definitions employ logical connectives, such as:

$$\text{NetworkNode} \sqsubseteq \exists \text{possesses.LogicalIdentifier}$$

indicating that every `NetworkNode` possesses at least one `LogicalIdentifier`. Here, $\sqcap$ denotes intersection (AND) and $\exists$ denotes existence, enabling precise expression of semantic constraints.

In summary, the ontology provides the logical and semantic structure, whilst the KG materialises this structure in concrete data. DL acts as the formal link between them, ensuring consistency, interpretability, and inferential capability. This triad — Ontology, KG, and DL — constitutes the methodological core of DT-II, enabling the II to be represented, monitored, and governed in a semantic and dynamic manner.

5 Conceptual Structure of the Digital Twin

This section presents a **reference architecture** for implementing the II Digital Twin as an **operational KG** (***ABox***) governed by an ***UO*** (***TBox***, ***RBox***). The aim is to make the model executable: to collect, normalise, validate, infer, and support decisions. The implementation objectives are:

(a) **Unified semantic inventory**: represent nodes, connections, and channels independently of technology;

(b) **Continuous synchronisation**: align the KG with telemetry and topology changes;

(c) **Detection and diagnosis**: identify inconsistencies, failures, and cascading impacts;

(d) **Verifiable governance**: encode and audit rules (SLA, jurisdiction, authorisation).

5.1 Reference Architecture

The architecture is organised into seven layers:

(i) **Ontological Layer (*TBox*)**: defines core classes (`NetworkNode`, `Connection`, `Channel`) and constraints;

(ii) **Identity Layer**: assigns stable IRIs to instances, separating identity from state;

(iii) **Ingestion Layer**: collects telemetry (YANG-Push, gNMI, SNMP, syslog) and normalises data into the Node–Connection–Channel triad;

(iv) **Persistence Layer (*ABox*, *RBox*)**: stores facts in RDF/OWL triplestores and maintains temporal history via observations;

(v) **Validation Layer**: applies structural constraints and logical consistency checks with DL reasoners;

(vi) **Reasoning Layer**: infers states (`DegradedConnection`, `SLAViolation`) and propagates failures;

(vii) **Exposure Layer**: provides SPARQL queries, dashboards, alerts, and integration with NOC/orchestration systems.

5.2 Minimal Data Model

The initial implementation focuses on three entities:

(a) **Node**: identity (`hasIdentifier`), operational state, timestamp;

(b) **Connection**: endpoints (`connectNode`), supported channel, metrics (latency, loss, jitter);

(c) **Channel**: medium type (fibre, copper, radio), capacity, QoS policy.

5.3 Integration of *ABox* and *RBox*

The ***ABox*** instantiates infrastructure elements (routers, links, channels) from inventory data (Claise, Clarke, and Lindblad 2019). The ***RBox***, based on $\mathcal{SROIQ}$ logic (Horrocks, Kutz, and Sattler 2006), governs relational properties:

(a) **Transitivity**: deduces cascading failures;

(b) **Mereology**: validates part–whole relations for SLA compliance (Effingham 2013, Ch. 8);

(c) **Property chains**: enable bidirectional navigation and anomaly tracing.

Together, ***ABox*** and ***RBox*** transform the KG into a computable model, enabling dynamic reasoning and governance.

5.4 Operational Flow

F1. Ingestion → normalisation → triple generation.
F2. Validation → integrity checks and anomaly detection.
F3. Inference → classification and rule-based derivations.
F4. Query/alert → recommendations or automated orchestration.

5.5 Example Scenario

In a fibre link failure:

(a) telemetry marks the `Channel` as unavailable;
(b) inference classifies the `Connection` as `InactiveConnection` and derives `SLAViolation`;
(c) redundancy is triggered (e.g., satellite backup).

In summary, the Digital Twin evolves into an **executable knowledge system**, where the ***TBox*** governs meanings and rules, and ***ABox/RBox*** capture state, validate consistency, and enable inference, transforming data into actionable decisions.

6 Use Cases

6.1 Use Case I: Ontological Compliance Regime and Logical Governance in DT-II

The Ontological Compliance Regime is the governance mechanism of **DT-II** that establishes authority and policy compliance over digital entities based on meaning, context, and relations — transcending traditional geophysical boundaries. In this model, jurisdiction ceases to be a static hardware attribute and becomes a **normative entity mediated by relators**.

6.1.1 Foundation and Axiomatisation

Defined in the ***TBox*** through restriction axioms, the Ontological Compliance Regime employs the Reasoner to determine which norms (e.g., LGPD, GDPR, SLAs) apply to real-time data flows. In the ***RBox***, property chains formalise propagation: if a `NetworkNode` is linked to a legal norm, all `Channels` and `Connections` mediated by relators inherit the same constraints.

6.1.2 Role of the Digital Twin

The Digital Twin acts as the guardian of jurisdiction. By ingesting telemetry (gNMI, YANG-Push, SNMP), the system populates the ***ABox***. Through **Semantic Elevation**, it verifies, for example, whether a health data packet traverses a node under a jurisdiction that does not meet the privacy requirements defined in the ***TBox***.

6.1.3 Operationalisation and Resilience

In dynamic events (e.g., DDoS attacks, BGP failures), the Ontological Compliance Regime guides the **MAPE-K** cycle: (a) **Analysis**: the reasoner detects violations of jurisdiction axioms; (b) **Planning**: the system seeks compliant alternatives; (c) **Execution**: configuration is applied via ***NETCONF*** only after logical validation.

In summary, the Ontological Compliance Regime transforms infrastructure into a ***juridically aware*** system, where compliance is mathematically verified by DL, ensuring governance integrity in complex and mutable networks.

6.2 Use Case II: Model Behaviour under a DDoS Attack

In a Distributed Denial of Service (DDoS) attack, response time is critical. The Digital Twin acts as a Semantic Defence System, leveraging the ***TBox***, ***RBox***, and ***ABox*** within the ***MAPE-K*** cycle as follows:

- **Monitoring** Telemetry (gNMI, YANG-Push, SNMP) updates the ***ABox*** in real time. For example, a traffic spike from 1 Gbps to 100 Gbps is recorded as:

```
inst:Unusual_Flow a ii:AnomalousFlow
```

- **Analysis** The Reasoner applies property chains from the ***RBox***: (a) the anomalous flow saturates `Channel_Alpha`; (b) transitivity propagates risk to dependent services (e.g., DNS); (c) the origin is traced to a specific **Ontological Compliance Regime**.
- **Planning** Consulting the ***TBox***, the system applies rules: if a critical service is at risk and the origin is external, mitigation (Blackholing or Rate Limiting) must occur at the edge. The plan prioritises nodes sustaining the network's **Ontological Normative Base**.
- **Execution** Logical decisions are translated into technical commands. Via ***NETCONF***, edge routers are reconfigured to filter malicious traffic.
- **Advantages of the Semantic Approach** Unlike conventional defences that block IPs, **DT-II** enables: (a) **Context Awareness**: cooperation agreements mapped in the ontology may trigger governance APIs instead of simple blocking; (b) **Mereological Resilience**: the Reasoner may sacrifice non-vital channels to preserve critical services; (c) **Traceability**: each mitigation step is explainable, linked to axioms in the ***TBox***; (d) **Compliance**: no action is executed without validation by the ***Ontological Normative Base***.
- **Sequence Flow** Mitigation occurs within milliseconds, moving from the **Active Infrastructure** to the **Logical Core** and back. Figure 3 illustrates the process.

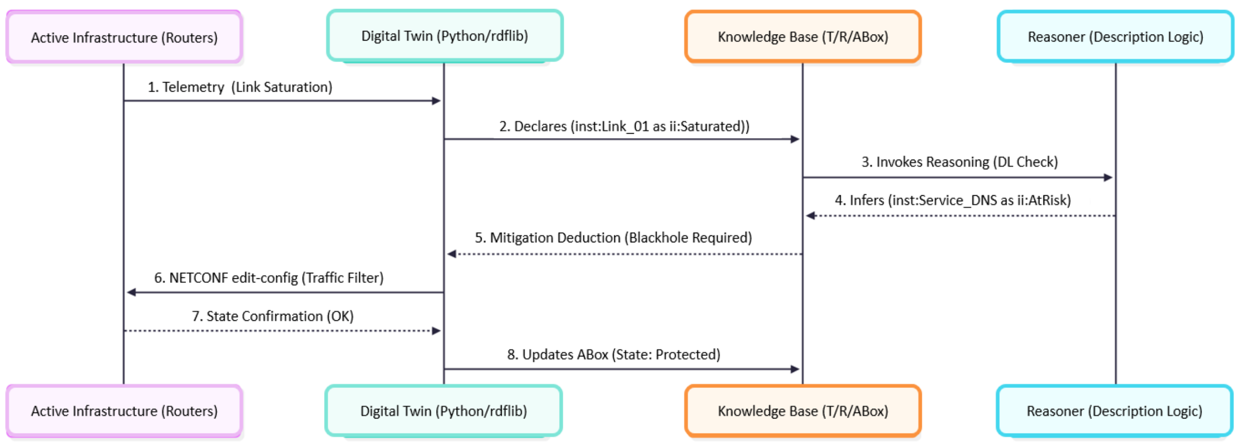

Figure 3: Sequence Diagram of DDoS Mitigation via DT-II

In summary, the ontology-based approach enables **policy-driven mitigation**, where assets of higher semantic and contractual value are prioritised, ensuring resilience and governance integrity.

7 Why the II Requires a Project like DT-II

The II must be understood not merely as cables and protocols, but as a **living system** that has already surpassed the limits of manual management. The **DT-II** project is necessary for four reasons: (R1) **From Static Management to Semantic Autonomy**: Current II management relies on rigid scripts and manual CLI commands. DT-II introduces **Logical Reasoning**, enabling the system to deduce solutions for complex failures and attacks under an ***Ontological Normative Base***; (R2) **NETCONF as an Interoperability Catalyst**: The growing adoption of ***NETCONF*** provides a vendor-independent, standardised channel for the Digital Twin to interact with the **Active Infrastructure**, updating ***ABox*** instances in real time; (R3) **Global Semantic Governance**: In a fragmented regulatory scenario (e.g., LGPD, GDPR), DT-II encodes laws and SLAs as axioms, ensuring compliance regardless of geographic routing complexity; (R4) **Universal Semantic Interoperability**: Grounded in the ***UO*** (that is, ***UFO***), DT-II resolves data silos, acting as a universal translator across domains (5G, Satellite, SD-WAN[4]), reducing costs and integration errors.

In summary, without DT-II, II remains an **Active Infrastructure** — processing packets without context. With DT-II, II gains **State Awareness**, evolving into an **Autonomic Network** that understands what it moves, why it moves, and whether it has logical permission to do so.

[4]Software-Defined Wide Area Network, a software-based approach to WAN management.

8 Conceptual Expansions of DT-II

This section deepens theoretical and practical aspects that complement the initial DT-II proposal, consolidating foundations and anticipating implementation challenges.

8.1 Inter-Domain Multi-Agent Coordination

Although the MAPE-K architecture is already described for internal agents, it is necessary to detail how multiple *Digital Twins* from different ASs interact. Inter-agent communication protocols such as FIPA-ACL or KQML may be employed for message exchange between graphs. In distributed attack scenarios (e.g., DDoS), planning agents from different ASs must negotiate routes and mitigation strategies, establishing consensus or federation among multiple MAPE-K cycles (Weyns et al. 2012).

8.2 Computational Scalability and *Stream Reasoning*

The use of expressive description logic (SROIQ) ensures semantic rigour but may generate bottlenecks in globally scaled ABoxes. To maintain millisecond-level response times, we propose the use of *Stream Reasoning* techniques and logical partitions of the ABox. This approach enables incremental reasoning over continuous data streams, ensuring DT-II operates in critical environments without compromising performance (Quin, Weyns, and Gheibi 2021; Andersson et al. 2023).

8.3 Conflict Resolution in the Ontological Normative Base

The Ontological Compliance Regime evaluates plans against strict rules, yet conflicts are inevitable in real infrastructures. For instance, the self-protection rule may clash with high-availability SLA requirements. Indeed, dealing with conflicting decisions and the adaptation actions they imply is recognised in the literature as one of the main open challenges for advancing self-adaptive systems (Quin, Weyns, and Gheibi 2021). To address this limitation and handle such situations, we suggest the use of defeasible logics[5] or ontological weight hierarchies. This approach enables the inference engine to resolve conflicting axioms in a consistent and transparent manner.

9 Conclusion and Future Work

This paper presented the conceptual foundations of **DT-II**, a multi-agent Digital Twin for II based on Ontology, Knowledge Graphs, and Description Logic. By formalising governance through the **Ontological Compliance Regime** and enabling policy-driven responses to complex events such as DDoS attacks, the model advances the vision of transforming II from a passive infrastructure into a **juridically aware autonomic system**. The proposed architecture demonstrates how ontological rigour can ensure resilience, compliance, and semantic interoperability in heterogeneous domains. All project results will be available at `https://github.com/LJuliaoBraga/dtii`.

Future work will focus on:

(i) simulation of the Digital Twin to evaluate alarms in ASs;

(ii) experimental validation of spatio-temporal response mechanisms;

(iii) integration with programmable infrastructures via ***NETCONF***;

(iv) exploration of semantic interoperability in domains such as 5G, satellite, and SD-WAN;

(v) participation in the evaluation of the two proposed telemetry alternatives — YANG-Push and gNMI — while remaining open to those that may emerge in the near future; and

(vi) Integration with Quantum Key Distribution (QKD) and Quantum Internet: We plan to explore the use of Quantum Communication channels for transmitting highly sensitive telemetry and governance metadata between the Active Infrastructure and the Digital Twin. As DT-II deals with critical "Semantic Jurisdictions" involving state-level privacy and strategic infrastructure topology, QKD integration will ensure synchronisation between the physical and logical layers remains secure even against future adversarial threats from quantum computing, maintaining the integrity of the "Ontological Normative Base" through a proven secure quantum-classical

[5] It is a logical system that admits conclusions based on rules that may be defeated by other rules or contrary evidence. In classical logic, once something is deduced, it cannot be withdrawn (monotonic reasoning). In defeasible logic, however, conclusions may be revised.

hybrid architecture (Kozlowski et al. 2023; Wang et al. 2024; Meter 2014). Future integration with Quantum Key Distribution (QKD) and Quantum Internet is a strategic differentiator of DT-II. We propose a hybrid architecture in which the Ontological Normative Base connects to quantum hardware to guarantee inviolable channels between the active infrastructure and the *Digital Twin*. This layer ensures that the MAPE-K cycle is not compromised by adversarial injections or tampering, reinforcing system resilience and security. By incorporating inter-domain multi-agent coordination, scalability via continuous synchronisation, normative conflict resolution, and quantum hybrid architecture, DT-II evolves from a conceptual proposal into an applicable engineering *framework*, capable of addressing the practical challenges of the Internet of the future.

In addition to the proposals listed above, it is worth highlighting the possibility of applying to DT-II the intelligent agents' architecture developed by Juliao Braga (2019) and previously explored in Julião Braga, Omar, and Granville (2015), and later in Juliao Braga, Stiubiener, and Henriques (2025). This integration may enhance the capacity for knowledge acquisition in restricted domains and strengthen the governance of ASs, contributing to greater efficiency, security, and explainability of the indicated actions, particularly in the application of the MAPE-K flow (Figure 2).

These steps aim to evolve the conceptual model design towards experimental deployment, culminating in its submission in 2026 as a draft to the IRTF ***nmrg*** within the CGI-IETF initiative, in collaboration with the following Brazilian universities: **UFABC** (SP), **UFRGS** (RS), **IFSPe** (PE), and **UNICAP** (PE).

10 Acknowledgements

The authors thank CGI.br for supporting the CGI-IETF Project, as well as the research groups and universities involved. Special thanks to colleagues and students who contributed with discussions, preliminary experiments, and ontological modelling activities.

Declaration on Generative AI

During the preparation of this work, the authors exclusively used Gemini and Copilot for grammar and spelling checks in Portuguese and English. Support was obtained for translations and relevant evaluations. All substantive content, analyses, and conclusions were produced by the authors, who reviewed and edited the text as necessary and assume full responsibility for the content of the publication.

References

Almeida, João Paulo A., Giancarlo Guizzardi, Tiago P. Sales, et al. (2019). *gUFO: A Lightweight Implementation of the Unified Foundational Ontology (UFO)*. http://purl.org/nemo/doc/gufo.

Almeida, João Paulo A., Giancarlo Guizzardi, Tiago Prince Sales, et al. (2026). *gUFO: A Gentle Foundational Ontology for Semantic Web Knowledge Graphs*. DOI: 10.48550/arXiv.2603.20948. arXiv: 2603.20948 [cs.AI]. URL: https://arxiv.org/abs/2603.20948.

Andersson, Jesper et al. (2023). "Architecting decentralized control in large-scale self-adaptive systems". In: *Computing* 105, pp. 1849–1882. DOI: 10.1007/s00607-023-01167-9.

Arp, Robert, Barry Smith, and Andrew D. Spear (2015). *Building Ontologies with Basic Formal Ontology*. 1st ed. Cambridge, MA, USA: The MIT Press. ISBN: 978-0-262-52781-1.

Baader, Franz et al. (2017). *Introduction to Description Logic*. Cambridge, UK: Cambridge University Press. ISBN: 9780521695428.

Braga, Juliao (2019). "Ambiente para aquisição de conhecimento por agentes em domínios restritos na infraestrutura da internet". Orientador: Nizam Omar. Membros da banca: Demi Getschko, Patrícia Takako Endo, Ismar Frango Siveira, Luciano Silva. Tese (Doutorado em Engenharia Elétrica e Computação). Em dupla titulação com o Instituto Superior Tecnico da Universidade de Lisboa. São Paulo: Universidade Presbiteriana Mackenzie, p. 128. URL: http://dspace.mackenzie.br/handle/10899/26511.

Braga, Juliao, Percival Henriques, et al. (2025). *Bases de Conhecimento*. 1a. Recife, PE, Brazil: Editora Publius. ISBN: 978-65-85007-43-6.

Braga, Juliao, Itana Stiubiener, and Percival Henriques (Nov. 2025). *Framework for Acquisition, Representation, and Use of Knowledge, Learning, and Collaboration among Intelligent Multi-Agent Systems in the Domain of Internet Infrastructure (SKAU)*. DOI: 10.31219/osf.io/qg36y_v3. URL: osf.io/qg36y_v3.

Braga, Julião, Nizam Omar, and Lisandro Granville (2015). "Uma proposta para o uso de Elementos Inteligentes em Domínios Restritos da Infraestrutura da Internet". In: *Anais do II Workshop Pré-IETF*. Recife: SBC, pp. 30–44. DOI: 10.5753/wpietf.2015.10467. URL: https://sol.sbc.org.br/index.php/wpietf/article/view/10467.

Claise, Benoit, Joe Clarke, and Jan Lindblad (2019). *Network programmability with YANG: the structure of network automation with YANG, NETCONF, RESTCONF, and gNMI*. Boston, MA, USA: Addison-Wesley Professional. ISBN: 978-0-13-518039-6.

Clemm, Alexander and Eric Voit (Sept. 2019). *Subscription to YANG Notifications for Datastore Updates*. Tech. rep. 8641. IETF. DOI: 10.17487/RFC8641. URL: https://www.rfc-editor.org/info/rfc8641.

Effingham, Nikk (2013). *An Introduction to Ontology*. English. Cambridge, UK; Malden, MA, USA: Polity Press, p. 224. ISBN: 978-0-7456-5254-2.

Grieves, Michael and John Vickers (2017). *Digital Twin: Manufacturing Excellence through Virtual Factory Replication*. NASA Technical Report NASA/TM-2017-219363. Disponível em: https://ntrs.nasa.gov. NASA.

Guerson, John et al. (2015). "OntoUML Lightweight Editor". In: *2015 19th IEEE International Enterprise Distributed Object Computing Conference (EDOC)*. Adelaide, Australia: IEEE, pp. 144–153. DOI: 10.1109/EDOC.2015.30. URL: https://doi.org/10.1109/EDOC.2015.30.

Guizzardi, Giancarlo (2005). *Ontological Foundations for Structural Conceptual Models*. CTIT Ph.D. Thesis Series. Enschede, The Netherlands: Universal Press. ISBN: 90-75176-81-3.

Guizzardi, Giancarlo, Claudenir M. Fonseca, et al. (2018). "Endurant types in ontology-driven conceptual modeling: Towards ontoUML 2.0". In: *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. Vol. 11157. Cham, Switzerland: Springer, pp. 136–150. ISBN: 978-3-030-00846-8. DOI: 10.1007/978-3-030-00847-5_12.

Guizzardi, Giancarlo and Nicola Guarino (2024). "Explanation, semantics, and ontology". In: *Data & Knowledge Engineering* 153, pp. 1–27. DOI: 10.1016/j.datak.2024.102325.

Hawkinson, Jon and Tony Bates (Mar. 1996). *Guidelines for creation, selection, and registration of an Autonomous System (AS)*. Tech. rep. 1930. RFC Editor. DOI: 10.17487/RFC1930. URL: https://www.rfc-editor.org/rfc/rfc1930.

Horrocks, Ian, Oliver Kutz, and Ulrike Sattler (2006). "The Even More Irresistible SROIQ". In: *Proceedings of the 10th International Conference on Principles of Knowledge Representation and Reasoning (KR)*. Lake District, United Kingdom: AAAI Press, pp. 57–67.

IBM Corporation (2005). *An Architectural Blueprint for Autonomic Computing*. https://www.ibm.com/autonomic/. Introduces the MAPE-K reference model (Monitor, Analyse, Plan, Execute, Knowledge).

Keet, C. Maria (2025). *An Introduction to Ontology Engineering*. Second. UK: College Publications. ISBN: 978-1-84890-020-2.

Kejriwal, Mayank, Craig A. Knoblock, and Pedro Szekely (2021). *Knowledge Graphs: Fundamentals, Techniques, and Applications*. Cambridge, Massachusetts, USA: MIT Press. ISBN: 9780262045094.

Kephart, Jeffrey O. and David M. Chess (2003). "The Vision of Autonomic Computing". In: *Computer*. Vol. 36. 1. IEEE, pp. 41–50. DOI: 10.1109/MCOM.2003.1160055.

Kozlowski, Wojciech et al. (Mar. 2023). *Architectural Principles for a Quantum Internet*. RFC 9340. DOI: 10.17487/RFC9340. URL: https://www.rfc-editor.org/info/rfc9340.

Landgrebe, Jobst and Barry Smith (2025). *Why Machines Will Never Rule the World: Artificial Intelligence without Fear*. 2nd ed. London and New York: Routledge. ISBN: 978-1-032-94140-0.

Meter, Rodney Van (2014). *Quantum Networking*. Hoboken, NJ: Wiley-ISTE, p. 360. ISBN: 978-1-84821-537-5.

Murch, Richard (2004). *Autonomic Computing*. Armonk, New York, USA: IBM Press, p. 336. ISBN: 978-0131440258.

Nardi, Daniele and Ronald J. Brachman (2010). "An Introduction to Description Logics". In: *Description Logic Handbook*. 2nd ed. Vol. 1. Cambridge, United Kingdom: Cambridge University Press, p. 40. ISBN: 978-0-521-15011-8.

Parks, Brandon A. (Sept. 2025). *An Introduction to Ontology Engineering: Foundations, Methods, and Practice. Build and Validate Knowledge Graphs*. English. ISBN: 9798267011358. Independently Published, p. 248.

Quin, Federico, Danny Weyns, and Omid Gheibi (2021). *Decentralized Self-Adaptive Systems: A Mapping Study*. DOI: 10.1109/SEAMS51251.2021.00014. arXiv: 2103.09074.

Thomas, Felipe et al. (July 2026). "Unified Network Abstraction Layer via Knowledge Graph". In: *Proceedings of the ACM/IRTF Applied Networking Research Workshop 2026 (ANRW'26)*. Held in conjunction with IETF 126. Vienna, Austria.

W3C OWL Working Group (Sept. 2012). *OWL 2 Web Ontology Language Document Overview (Second Edition)*. W3C Recommendation. URL: https://www.w3.org/TR/owl2-overview/.

Wang, Chonggang et al. (June 2024). *Application Scenarios for the Quantum Internet*. RFC 9583. DOI: 10.17487/RFC9583. URL: https://www.rfc-editor.org/info/rfc9583.

Weyns, Danny et al. (2012). "On Patterns for Decentralized Control in Self-Adaptive Systems". In: *Software Engineering for Self-Adaptive Systems II*. Vol. 7475. LNCS. Springer, pp. 76–107.

Zhou, Cheng et al. (Feb. 2026). *Network Digital Twin Concept*. Internet-Draft draft-irtf-nmrg-network-digital-twin-arch-12. Work in Progress. Internet Engineering Task Force (IETF).